

Project Report: Design and Implementation of a 5V Regulated Power Supply

Author: Pouria Saeedi

Date: July 6, 2026

Course: Embedded Systems / Power Electronics Lab

1. Introduction & Applications

This project presents the design, simulation, and PCB layout of a regulated 5V DC power supply. This circuit serves as a fundamental building block for powering microcontrollers (e.g., AVR, STM32, Arduino), sensor nodes, development boards, and low-power digital systems that require a stable, low-noise supply voltage to function reliably.

2. Circuit Architecture and Working Principle

The system converts high-voltage AC mains power to a stable, regulated 5V DC output through four primary stages:

1. **Voltage Step-Down:** The 220V AC mains input (at 50 Hz) is stepped down using an external step-down transformer to a safe low-voltage AC level (typically 9V RMS).
2. **Bridge Rectification:** A full-wave bridge rectifier converts the sinusoidal AC voltage into pulsating DC.
3. **Filtering Stage:** A 470 μ F bulk electrolytic capacitor (C_1) filters out the low-frequency voltage ripple, providing a smoother DC input for the regulator.
4. **Voltage Regulation:** The LM1084IS-5.0 Low Dropout (LDO) regulator steps down and stabilizes the voltage to a precise 5V DC output.

5. Decoupling Stage: A combination of a 10 μ F tantalum capacitor and a 100nF ceramic capacitor at the output filters high-frequency noise and ensures transient stability under varying load currents.
-

3. Technical Specifications

- Input Voltage (Mains): 220V AC RMS, 50 Hz
 - Rectified Peak Voltage (V_{peak}): ~11.2V DC (after diode drop)
 - Bulk Filtering Capacity: 470 μ F
 - Regulator Model: LM1084IS-5.0 (TO-263 package)
 - Output Voltage: 5.00V DC
 - Maximum Output Current: Up to 5A (limited by thermal dissipation and trace widths)
-

4. PCB Design Specifications

The physical PCB was designed in Altium Designer with the following design rules:

- Power Traces: 40 mil to 60 mil trace widths to handle high current loads and minimize resistive losses.
- Signal/Feedback Traces: 20 mil width.
- Grounding Plane: A dedicated solid Polygon Pour on the bottom layer to provide a low-impedance ground return path and aid in heat dissipation.
- Component Placement: Decoupling capacitors (100nF) were placed as close as possible to the regulator pins to minimize parasitic inductance.